Graph learning for continuous Lyapunov models: lasso, MCP, SCAD, adaptive lasso and the dense start

Simulation results

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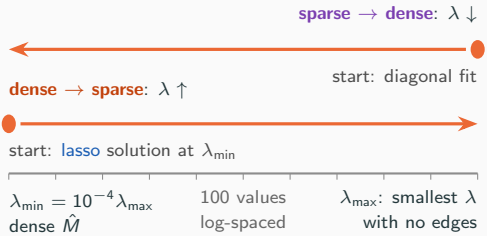
Model and estimators

GCLM $M\Sigma + \Sigma M^\top + C = 0$, M stable; graph: edge $j \to i \iff M_{ij} \neq 0 \quad (i \neq j)$

Every estimator: $\hat{M}_\lambda \in \arg\min_M \; L(M) + \sum_{i \neq j} P_\lambda\big(|M_{ij}|\big)$, for 100 values of λ; diagonal of M unpenalised

- **direct loss** [1]: $L(M) = \frac{1}{2}\big\|M\hat{\Sigma} + \hat{\Sigma}M^\top + C\big\|_F^2$
- **log-likelihood loss** [2]: $L(M) = \log\det\Sigma(M) + \operatorname{tr}\big(\Sigma(M)^{-1}\hat{\Sigma}\big)$, $\Sigma(M)$ from the GCLM equation

- **lasso**: $P_\lambda(t) = \lambda t$
- **MCP** [5], $\gamma = 3$: $P_\lambda(t) = \begin{cases} \lambda t - t^2/(2\gamma), & t \leq \gamma\lambda \\ \gamma\lambda^2/2, & t > \gamma\lambda \end{cases}$
- **SCAD** [4], $\gamma = 3.7$
- **adaptive lasso** [6], convex: $P_\lambda(|M_{ij}|) = \lambda\, w_{ij}\, |M_{ij}|$
  $w_{ij} = \max_{kl} |\tilde{M}_{kl}| \,/\, |\tilde{M}_{ij}|$, $\tilde{M}$: lasso solution at $\lambda_{\min}$



Computing a path: the 100 problems are solved one after another in the direction of the arrow.
Each solve (FISTA [9]) is initialised with the solution of the previous one.

Data, and from a path to one graph

True models [1, §5]: $M_{ij} = \omega_{ij}\varepsilon_{ij}$, $\omega_{ij} \sim \mathrm{Bern}(k/p)$, $\varepsilon_{ij} \sim N(0,1)$; $M_{ii} = -\sum_{j\neq i}|M_{ij}| - |\varepsilon_{ii}|$
$p = 10, 20$ (and 15 to 50), $n = 10^3, 10^4$ or ∞, standardised; 100 graphs per p: $k = 1, \ldots, 4$, 25 each, averaged
True C [1, Fig. 5]: $2I$ on the main slides, others on the backup slides; C **in the fit**: $2I$ or rescaled

A path gives 100 graphs, one per λ.
Two ways to choose one of them:

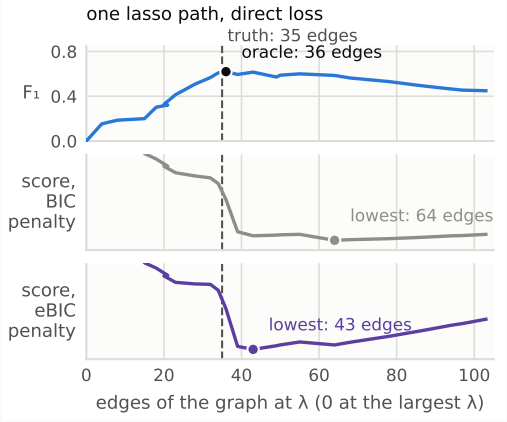
- **oracle**: $\hat{\lambda} = \arg\max_\lambda F_1(\hat{M}_\lambda)$ uses the truth
- **score**: the support S on the path with the lowest

$$\mathrm{score}(S) = \underbrace{n\left[\log\det\Sigma_S + \mathrm{tr}(\Sigma_S^{-1}\hat{\Sigma})\right]}_{\text{loss}} + \underbrace{\mathrm{pen}(S)}_{\text{penalty}}$$

M_S: least-squares refit of the direct loss on S
Σ_S: solves $M_S\Sigma_S + \Sigma_S M_S^\top + C = 0$

  - **BIC**: $\mathrm{pen}(S) = (p + |S|)\log n$
  - **eBIC** [1]: $\mathrm{pen}(S) = (p + |S|)\log n + 4|S|\log p$

Measured: directed F_1 of the chosen graph; AUPR of the path



Why rescale C?

Standardising, $X \to D^{-1}X$ with $D = \mathrm{diag}(s_1, \ldots, s_p)$, is a change of variables:

$$M\Sigma + \Sigma M^\top + C = 0 \quad \Longleftrightarrow \quad \tilde{M}R + R\tilde{M}^\top + \tilde{C} = 0$$

$\tilde{M} = D^{-1}MD, \quad \tilde{C} = D^{-1}CD^{-1}, R = D^{-1}\Sigma D^{-1}$

Data generated with $C = 2I$ therefore follow, after standardising, the model with

$$\tilde{C} = 2\,\mathrm{diag}\big(1/s_1^2, \ldots, 1/s_p^2\big) \qquad \text{(the rescaled } C\text{)}$$

Findings

Direct loss: $p = 10, 20$; $n = 10^3, 10^4, \infty$; true $C = 2I$; C in the fit $= 2I$ or rescaled

V1 Sparse $\rightarrow$ dense MCP and SCAD "lose" to the lasso.
They lock in wrong edge directions early on the path.

V2 Dense $\rightarrow$ sparse MCP and SCAD better than lasso, by more with the rescaled C and as p and n grow.

V3 The convex adaptive lasso, from the same dense start, does as well as MCP and SCAD, with both C.
So the dense start matters, not the nonconvexity.

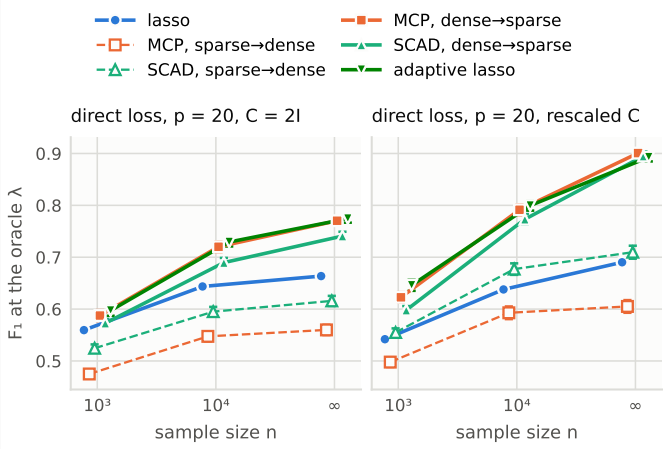
V4 With the rescaled C the BIC penalty selects too many edges from $p = 20$ on, and the eBIC penalty is better from $p = 30$ on.
With $C = 2I$ the eBIC penalty drops true edges and is worse at every p.

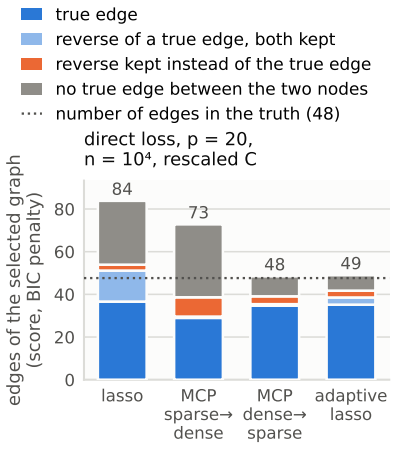
V5 The greedy search from random graphs needs manny starting graphs (e.g., $r = 300$) to find a good solution.

Log-likelihood loss: $p = 10, 20$; $n = 10^3, 10^4$; both C

V6 The findings of the direct loss transfer, if the dense start is the lasso's dense end.
Started from the exact fit $-\frac{1}{2} C \hat{\Sigma}^{-1}$ instead, dense $\rightarrow$ sparse MCP does not beat the lasso.

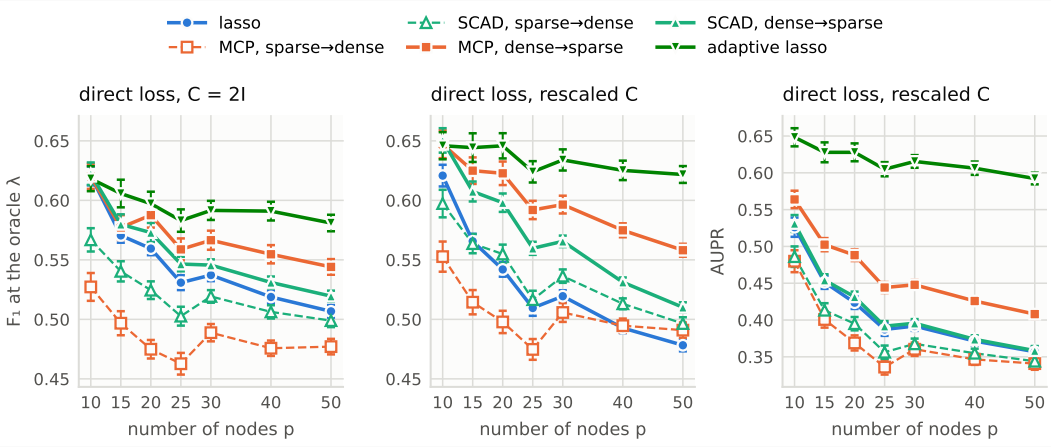
V1–V3, direct loss: the two path orders and the adaptive lasso





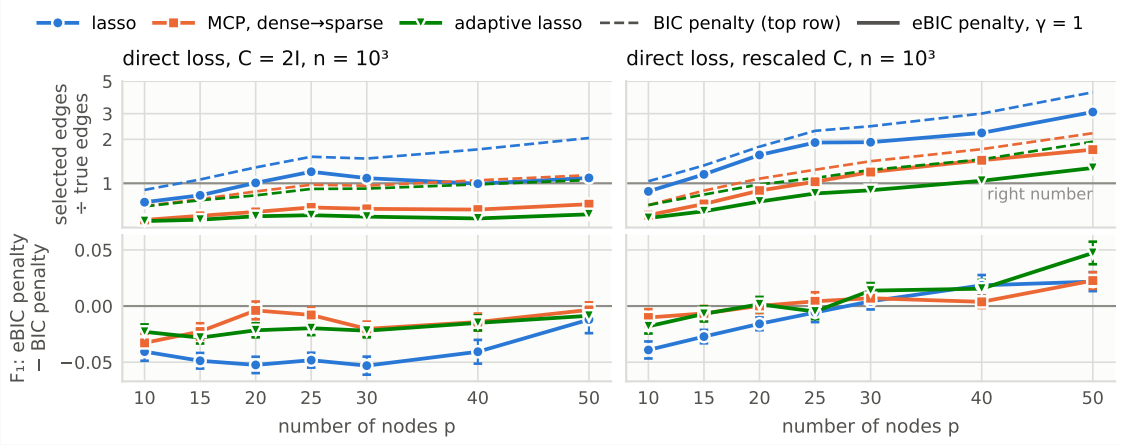
Left: dashed (sparse → dense) lies below the lasso; solid (dense → sparse) lies above.
Right: sparse → dense MCP often keeps the reverse of a true edge instead of the edge (orange).

V2–V3, direct loss: the advantage grows with p ($n = 10^3$)



The lasso degrades with p; the adaptive lasso barely does.

V4, direct loss: the BIC penalty against Dettling's eBIC penalty ($n = 10^3$)



The eBIC term $4\gamma \log p$ per edge raises F_1 only where the BIC penalty gives too many edges: with the rescaled C from $p = 30$ on.

V5, direct loss: the greedy search

Score of a graph G: fit M on G by least squares on the direct loss ([3]: maximise the likelihood), then $\mathrm{score}(G)$ as on slide 2, with the BIC penalty.

Search from a graph S, as in [3]:

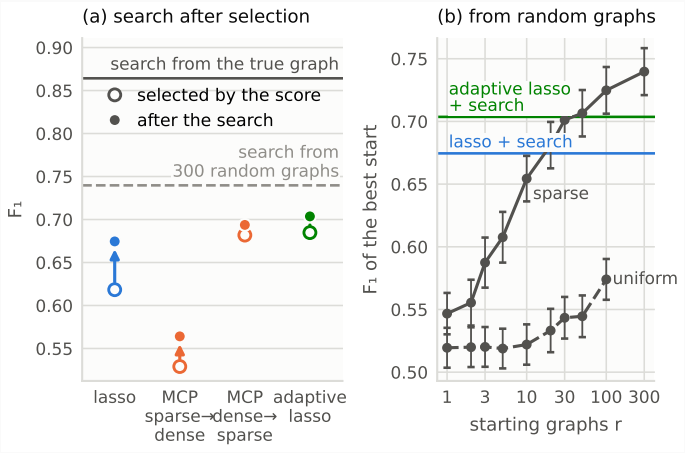
1. score every graph that differs from S by one edge (added, deleted, reversed)
2. if the best of them has a lower score than S, it becomes S; repeat

Where the search starts

(a) the graph the score selects on a path (lasso, MCP/SCAD, adaptive lasso)

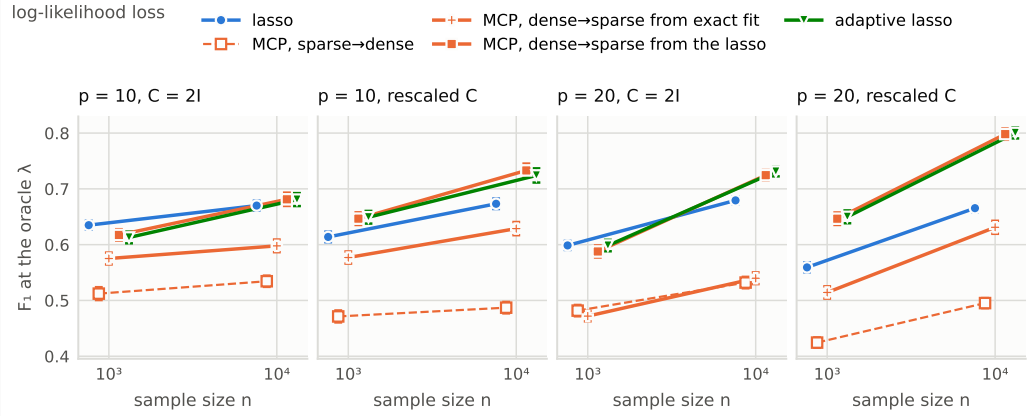
(b) r random graphs and the empty graph ([3]: $r = 300$)

Random graphs: sparse, edge probability from $U[0, 0.3]$; uniform, $1/2$ per edge [8].



$p = 10$, $n = 10^4$, rescaled C, true $C = 2I$, direct loss.

V6, log-likelihood loss: the dense start from the lasso wins here too



Rescaled C:

Filled squares (MCP from the lasso's dense end) and green (adaptive lasso) lie above the lasso, by more at $p = 20$.

Crosses (MCP from the exact fit $-\frac{1}{2}C\hat{\Sigma}^{-1}$ [2]) do not.

Hollow squares (sparse → dense) lose everywhere.

Next steps

- **Running on the cluster**
  - For $n = 10^3$: every stored path rescored with the likelihood refit M_S (instead of the least-squares refit).
- **Open:** a theory behind sparse $\rightarrow$ dense or adaptive lasso for GCLM? Next simulations?

References

[1] Dettling, Drton & Kolar (2024). On the lasso for graphical continuous Lyapunov models. *CLeaR 2024*.

[2] Varando & Hansen (2020). Graphical continuous Lyapunov models. *UAI 2020*.

[3] Améndola, Dettling, Drton, Onori & Wu (2020). Structure learning for cyclic linear causal models. *UAI 2020*.

[4] Fan & Li (2001). Variable selection via nonconcave penalized likelihood and its oracle properties. *JASA* 96.

[5] Zhang (2010). Nearly unbiased variable selection under minimax concave penalty. *Ann. Statist.* 38.

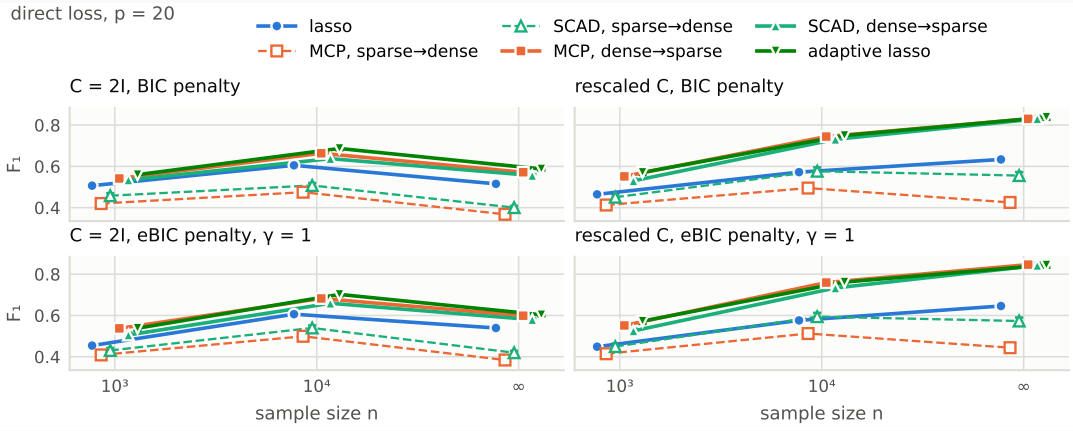
[6] Zou (2006). The adaptive lasso and its oracle properties. *JASA* 101.

[7] Chen & Chen (2008). Extended Bayesian information criteria for model selection with large model spaces. *Biometrika* 95.

[8] Nowzohour, Maathuis, Evans & Bühlmann (2017). Distributional equivalence and structure learning for bow-free acyclic path diagrams. *Electron. J. Stat.* 11.

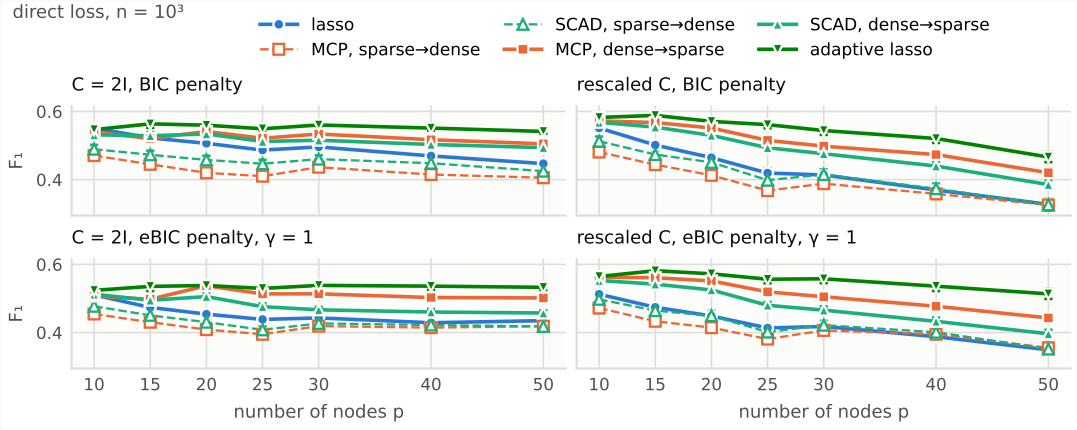
[9] Beck & Teboulle (2009). A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM J. Imaging Sci.* 2.

Backup: V1–V3 with the graph the score selects ($p = 20$)



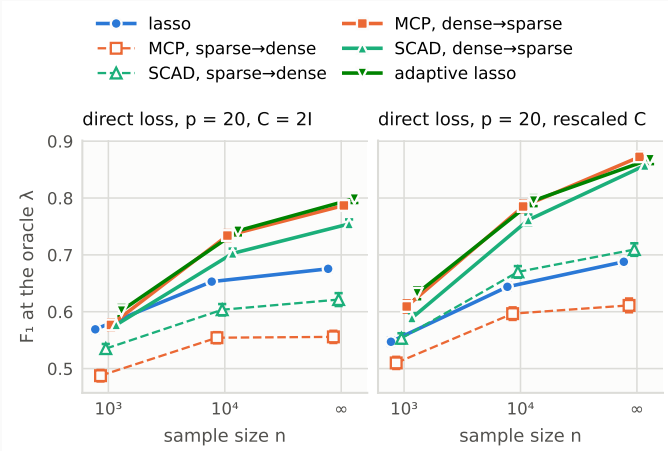
F_1 of the graph the score selects, instead of at the oracle λ; true $C = 2I$; at $n = \infty$ the score uses $n = 10^6$.
Same ordering as at the oracle: sparse $\rightarrow$ dense below or level with the lasso; dense $\rightarrow$ sparse and the adaptive lasso above it, by as much or more.

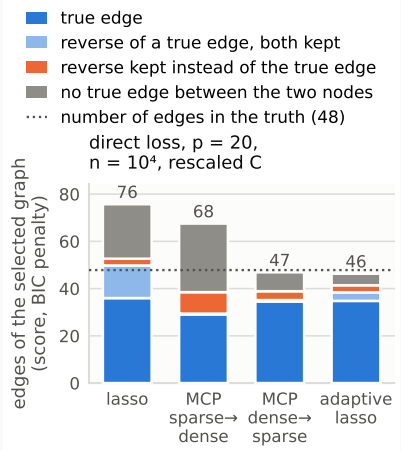
Backup: V2–V3 with the graph the score selects, over p ($n = 10^3$)



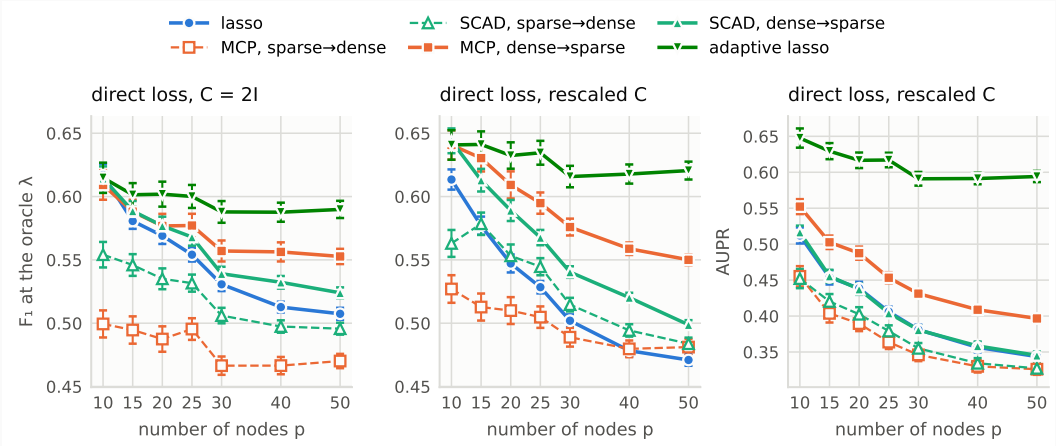
As at the oracle λ, the gain of the dense start over the lasso grows with p; with either penalty it is larger.

Backup, true C diagonal, entries from $U[2, 4]$: V1–V3, direct loss

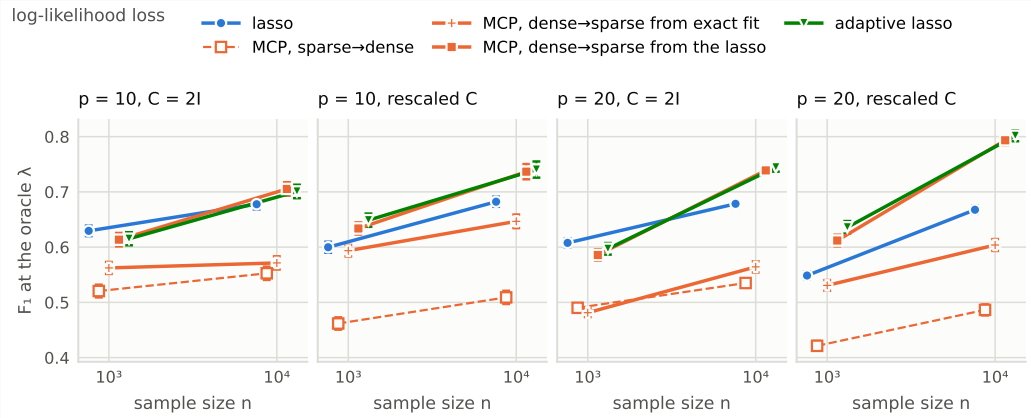




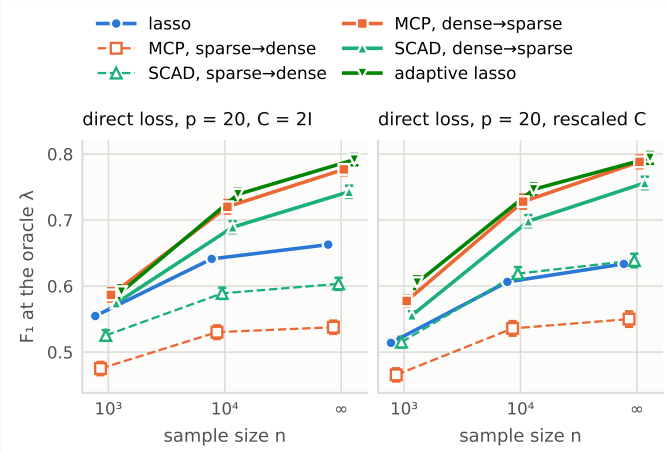
Backup, true C diagonal, entries from $U[2,4]$: V2–V3, direct loss, over p ($n = 10^3$)

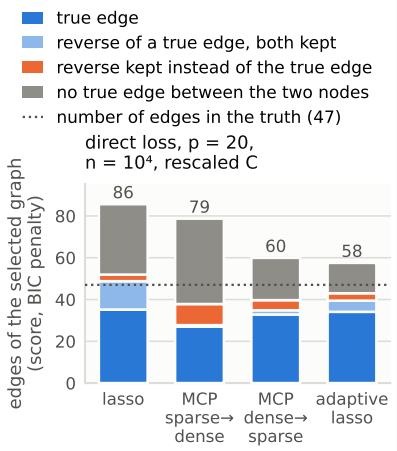


Backup, true C diagonal, entries from $U[2, 4]$: V6, log-likelihood loss

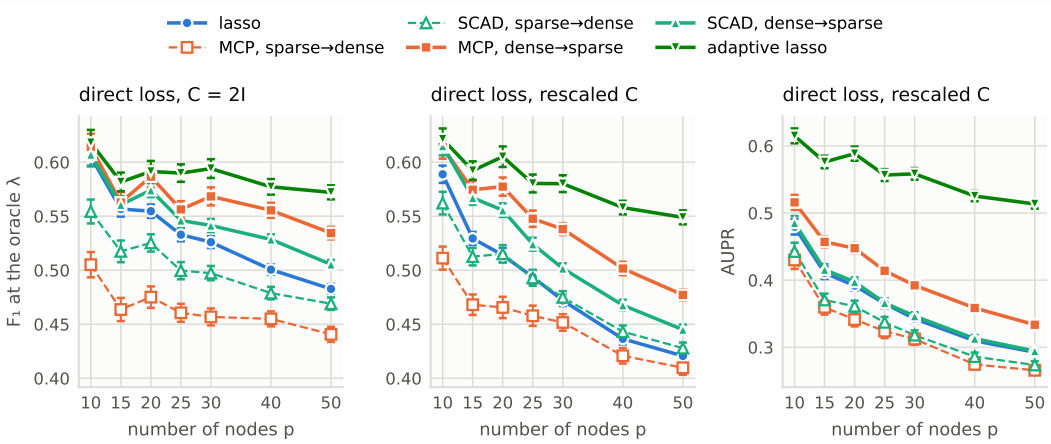


Backup, true C diagonal, entries from $U[0.5, 4]$: V1–V3, direct loss

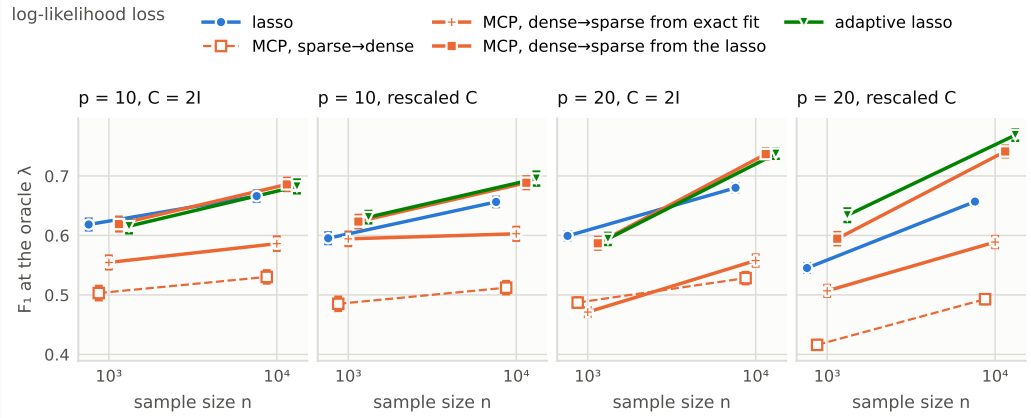




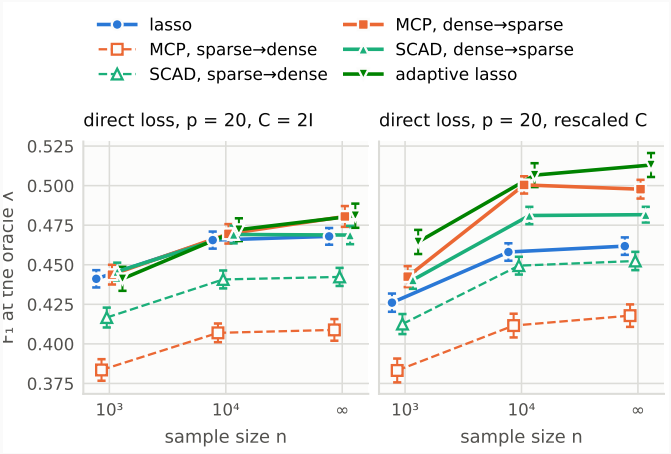
Backup, true C diagonal, entries from $U[0.5, 4]$: V2–V3, direct loss, over p ($n = 10^3$)

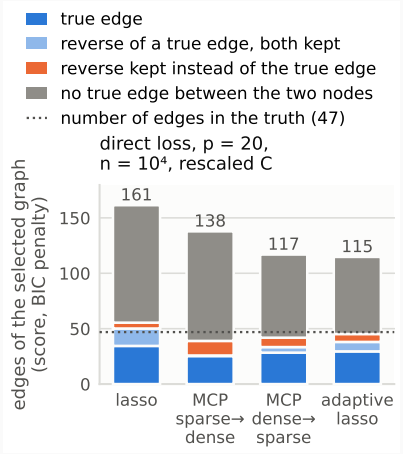


Backup, true C diagonal, entries from $U[0.5, 4]$: V6, log-likelihood loss

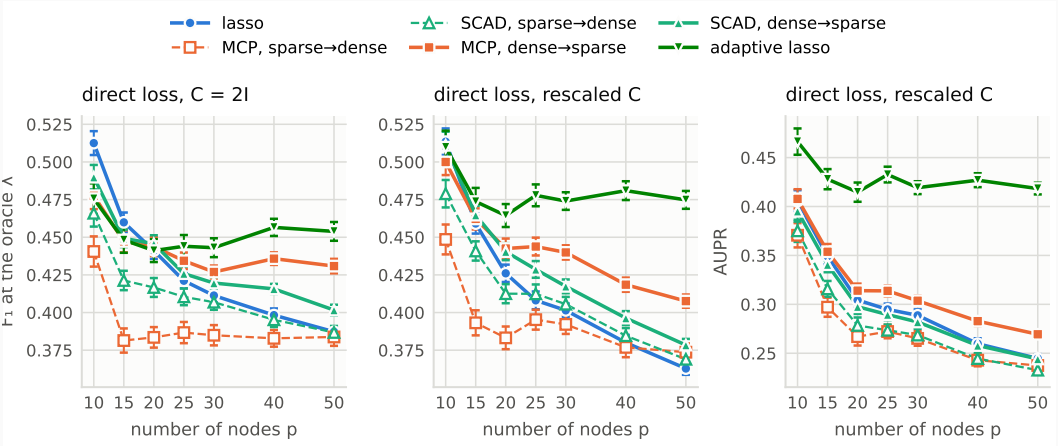


Backup, true C non-diagonal: V1–V3, direct loss





Backup, true C non-diagonal: V2–V3, direct loss, over p ($n = 10^3$)



Backup, true C non-diagonal: V6, log-likelihood loss

